

BUCURESTI AFUMATI, RomaniaWMO: **154210**Lat: **44.5000N**Lon: **26.2128E**Elev: **91**StdP: **100.23**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-10.2	-15.9	0.9	-12.0	-12.8	1.3	-9.4	14.6	-1.1	11.9	-0.8	2.5	240	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	34.1	21.3	32.6	21.2	31.0	20.5	23.3	30.3	22.4	29.5	21.5	28.5	2.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	16.0	25.3	20.2	15.0	24.6	19.2	14.1	23.9	69.5	30.2	66.1	29.4	63.1	28.9	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.0	6.8	-17.5	36.5	3.6	2.0	-20.1	37.9	-22.2	39.1	-24.2	40.2	-26.8	41.6		
(5)			-17.7	25.1	3.5	1.2	-20.3	25.9	-22.4	26.6	-24.4	27.3	-26.9	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-1.2	1.4	6.4	12.0	17.5	21.4	23.4	23.3	18.1	12.0	6.4	0.6
	DBStd	9.47	4.75	5.27	4.62	3.64	3.27	3.03	2.59	2.80	3.44	4.00	4.26	4.50
	HDD10.0	1177	348	241	128	21	1	0	0	0	0	25	122	291
	HDD18.3	2857	607	473	371	190	56	8	1	2	45	199	357	549
	CDD10.0	1846	0	2	15	82	232	343	416	411	244	87	15	0
	CDD18.3	485	0	0	0	2	28	101	158	154	39	3	0	0
	CDH23.3	5170	0	0	1	46	358	1043	1652	1628	401	41	1	0
	CDH26.7	2047	0	0	0	9	85	391	721	723	113	6	0	0
Wind	WSAvg	2.8	2.9	3.2	3.4	3.2	2.8	2.5	2.4	2.4	2.6	2.5	2.7	2.9
Precipitation	PrecAvg	631	41	35	40	49	69	77	64	54	54	44	47	47
	PrecMax	1077	126	99	113	160	233	207	222	146	270	132	215	171
	PrecMin	374	3	2	3	1	8	10	5	1	4	0	1	0
	PrecStd	141	27	23	28	29	46	46	43	37	47	35	34	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	18.0	22.1	28.0	31.1	34.2	36.9	36.4	32.1	27.1	21.2	13.5
		MCWB	8.1	11.6	12.6	16.5	19.8	22.0	21.6	21.3	19.8	17.4	14.8	9.5
	2%	DB	8.9	14.2	19.6	24.1	29.0	32.2	34.1	34.1	29.8	24.0	17.9	10.2
		MCWB	6.5	9.0	11.4	15.1	18.5	21.6	21.3	21.2	18.9	16.4	13.2	8.0
	5%	DB	6.2	11.1	17.0	21.5	27.1	30.8	32.2	32.5	27.8	21.1	14.9	8.0
		MCWB	5.1	7.5	10.2	13.8	17.8	20.9	21.3	20.9	18.2	15.2	11.8	6.5
	10%	DB	4.6	8.2	14.6	19.3	25.1	29.0	30.8	30.9	25.8	18.8	12.2	6.1
		MCWB	3.7	6.1	8.9	12.6	17.0	20.1	20.8	20.3	17.3	13.8	10.3	5.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	11.6	13.9	17.5	21.2	24.0	24.9	25.0	21.2	19.3	15.5	10.8
		MCDB	11.9	17.0	20.4	24.9	28.7	31.3	30.3	30.5	28.3	25.3	20.0	12.3
	2%	WB	6.8	9.7	12.0	16.0	19.8	22.6	23.2	23.3	19.9	17.3	13.7	8.2
		MCDB	8.1	13.5	18.2	21.9	26.7	29.9	30.2	30.2	26.6	22.1	17.1	9.9
	5%	WB	5.2	8.0	10.6	14.8	18.7	21.7	22.4	22.0	19.0	15.9	12.2	6.7
		MCDB	6.3	10.5	15.9	20.2	24.8	28.8	29.3	29.2	25.5	20.0	14.5	7.8
	10%	WB	3.6	6.3	9.4	13.6	17.7	20.9	21.5	21.1	18.1	14.7	10.9	5.2
		MCDB	4.3	8.1	13.7	18.2	23.6	27.4	28.3	28.4	24.1	18.1	12.3	6.0
Mean Daily Temperature Range	5% DB	MDBR	6.8	8.5	10.6	11.6	12.4	12.1	12.5	12.9	11.9	10.4	8.1	6.6
		MCDBR	9.6	13.6	15.8	15.6	15.6	14.8	15.2	15.5	15.4	14.6	12.9	9.8
		MCWBR	7.2	8.9	8.8	7.7	6.9	5.9	5.5	5.3	6.3	7.8	8.2	7.3
	5% WB	MCDBR	9.2	12.4	14.4	13.6	13.9	13.7	13.3	13.5	13.3	13.1	11.0	9.0
		MCWBR	7.1	8.4	8.5	7.2	6.9	5.9	5.7	5.6	6.2	7.5	7.4	7.0
Clear-Sky Solar Irradiance	taub	0.346	0.378	0.413	0.450	0.425	0.434	0.445	0.459	0.423	0.400	0.374	0.350	
	taud	2.278	2.194	2.157	2.081	2.229	2.236	2.216	2.170	2.237	2.269	2.295	2.291	
	Ebn at Noon	747	787	811	812	848	839	824	796	793	754	708	699	
	Edh at Noon	85	111	131	154	137	136	138	139	120	102	84	77	
All-Sky Solar Radiation	RadAvg	1.37	2.21	3.43	4.70	5.88	6.34	6.43	5.74	4.16	2.70	1.47	1.13	
	RadStd	0.21	0.34	0.37	0.48	0.43	0.44	0.42	0.31	0.31	0.33	0.19	0.17	